

JAVA BANK ACCOUNT SYSTEM

1. AIM

To design and implement a Bank Account System in Java using object-oriented programming concepts such as classes, objects, attributes, methods, constructors, inheritance, method overriding, deposits, withdrawals, interest calculation, and fund transfers.

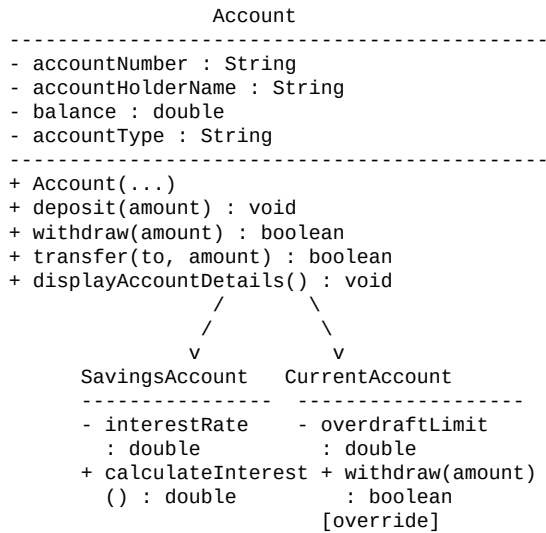
2. DESCRIPTION

The Bank Account System contains a base class named Account. It stores account number, account holder name, balance, and account type. The Account class provides deposit(), withdraw(), transfer(), and displayAccountDetails() methods. SavingsAccount extends Account and adds interestRate with calculateInterest(). CurrentAccount also extends Account and adds overdraftLimit. It overrides withdraw() to allow withdrawal up to the permitted overdraft limit. Objects of both account types are created and transactions are performed.

3. ALGORITHM

1. Start the program.
2. Create the Account base class.
3. Declare account number, holder name, balance, and account type.
4. Create a constructor to initialize account details.
5. Implement deposit() to add money to the balance.
6. Implement withdraw() to withdraw money when sufficient balance is available.
7. Implement transfer() to move money from one account to another.
8. Implement displayAccountDetails() to show account information.
9. Create SavingsAccount by inheriting Account and add interestRate.
10. Implement calculateInterest() for the savings account.
11. Create CurrentAccount by inheriting Account and add overdraftLimit.
12. Override withdraw() in CurrentAccount to support the overdraft limit.
13. Create SavingsAccount and CurrentAccount objects.
14. Display details before transactions.
15. Perform deposits, withdrawals, interest calculation, and fund transfer.
16. Display final account details and balances.
17. Stop the program.

4. UML DIAGRAM



Inheritance relationship:
SavingsAccount → Account
CurrentAccount → Account

5. SOURCE CODE WITH COMMENTS

// BankAccountSystem.java

```
class Account {
    protected String accountNumber;
    protected String accountHolderName;
    protected double balance;
    protected String accountType;

    Account(String accountNumber, String accountHolderName,
            double balance, String accountType) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
        this.accountType = accountType;
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.printf("Deposited Rs. %.2f into %s Account.%n",
                    amount, accountType);
        }
    }

    boolean withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.printf("Withdrew Rs. %.2f from %s Account.%n",
                    amount, accountType);
            return true;
        }
        System.out.println("Insufficient balance.");
        return false;
    }

    boolean transfer(Account to, double amount) {
        if (withdraw(amount)) {
            to.deposit(amount);
            return true;
        }
        return false;
    }

    void displayAccountDetails() {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Account Holder : " + accountHolderName);
        System.out.println("Account Type : " + accountType);
        System.out.printf("Balance : Rs. %.2f%n", balance);
    }

    double getBalance() {
        return balance;
    }
}
```

```

    }
}

class SavingsAccount extends Account {
    private double interestRate;

    SavingsAccount(String no, String name, double balance,
        double interestRate) {
        super(no, name, balance, "Savings");
        this.interestRate = interestRate;
    }

    double calculateInterest() {
        double interest = balance * interestRate / 100;
        balance += interest;
        System.out.printf("Interest calculated: Rs. %.2f%n", interest);
        return interest;
    }
}

class CurrentAccount extends Account {
    private double overdraftLimit;

    CurrentAccount(String no, String name, double balance,
        double overdraftLimit) {
        super(no, name, balance, "Current");
        this.overdraftLimit = overdraftLimit;
    }

    @Override
    boolean withdraw(double amount) {
        if (amount > 0 && amount <= balance + overdraftLimit) {
            balance -= amount;
            System.out.printf("Withdrew Rs. %.2f from Current Account.%n",
                amount);
            return true;
        }
        System.out.println("Overdraft limit exceeded.");
        return false;
    }
}

public class BankAccountSystem {
    public static void main(String[] args) {
        SavingsAccount savings =
            new SavingsAccount("SA305", "Anjali", 12000, 1.5);

        CurrentAccount current =
            new CurrentAccount("CA408", "Kiran", 7000, 3000);

        System.out.println("===== BEFORE TRANSACTIONS =====");
        System.out.println("Savings Account");
        savings.displayAccountDetails();

        System.out.println("\nCurrent Account");
        current.displayAccountDetails();

        System.out.println("\n----- TRANSACTIONS -----");

        savings.deposit(3000);
        savings.withdraw(2000);
        savings.calculateInterest();
        savings.transfer(current, 3500);
        current.withdraw(8500);

        System.out.println("\n===== FINAL DETAILS =====");
        System.out.printf("Savings Account Final Balance : Rs. %.2f%n",
            savings.getBalance());
        System.out.printf("Current Account Final Balance : Rs. %.2f%n",
            current.getBalance());
    }
}

```

6. OUTPUT SCREENSHOT

```
===== BEFORE TRANSACTIONS =====
Savings Account
Account Number : SA305
Account Holder : Anjali
Account Type : Savings
Balance : Rs. 12000.00

Current Account
Account Number : CA408
Account Holder : Kiran
Account Type : Current
Balance : Rs. 7000.00

----- TRANSACTIONS -----
Deposited Rs. 3000.00 into Savings Account.
Withdrew Rs. 2000.00 from Savings Account.
Interest calculated: Rs. 195.00
Withdrew Rs. 3500.00 from Savings Account.
Deposited Rs. 3500.00 into Current Account.
Withdrew Rs. 8500.00 from Current Account.

===== FINAL DETAILS =====
Savings Account Final Balance : Rs. 9695.00
Current Account Final Balance : Rs. 2000.00
```

7. RESULT

The Bank Account System was successfully designed and implemented in Java. The program demonstrates attributes, methods, constructors, object creation, inheritance, method overriding, deposits, withdrawals, interest calculation, and fund transfers. The final balances are displayed successfully.